



Strengthening the Policy-Research Nexus to Address SDG Trade-offs and Synergies in Climate-Smart Agriculture in Kenya

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Abstract

This study examines the integration of Climate-Smart Agriculture (CSA) within Kenya's national and county-level policy frameworks over the past decade, with a focus on alignment with the Sustainable Development Goals (SDGs) and the effectiveness of the research-policy interface. A desk-based review of 96 policy documents was combined with a bibliometric analysis of 1,307 publications (2015–2025) to assess patterns in policy design, implementation systems, and knowledge production. The findings show that CSA is widely embedded across policy frameworks, with many documents providing actionable guidance on practices such as sustainable grazing, water management, improved livestock systems, and early warning mechanisms. Alignment is strongest with SDG 13 (Climate Action) and SDG 2 (Zero Hunger), while SDG 5 (Gender Equality) and SDG 15 (Life on Land) are less consistently operationalized. Although policy frameworks increasingly recognize synergies and trade-offs, implementation remains uneven, particularly in relation to equity and indigenous knowledge. Institutional analysis indicates strong vertical coordination and clarity of roles, but financing, enforcement, and accountability mechanisms remain weak. Evidence use is inconsistent, with many policies relying on non-systematic inputs despite referencing learning frameworks. Bibliometric results show a sharp increase in CSA research output, rising from 22 publications in 2015 to over 215 in 2025. However, research leadership remains dominated by international institutions, with limited local capacity. Overall, the study highlights the need to strengthen implementation systems, enhance local research leadership, and better align evidence with policy to achieve sustainable CSA outcomes.

Key Words: Climate-Smart Agriculture (CSA), Research-policy interface, Sustainable Development Goals (SDGs), Agricultural policy, Evidence-based policymaking, Kenya

1. Introduction

Agriculture remains central to Kenya's economy and livelihoods, underpinning food security, employment, and rural development (Alawode, 2025; Ng'alu, 2025). In recent years, Climate-Smart Agriculture (CSA) has gained prominence as a strategic framework for addressing the interconnected challenges of climate change, agricultural productivity, and environmental sustainability (Ngigi & Muange, 2022; Onyango et al., 2025; Waaswa et al., 2024). Therefore, by simultaneously aiming to enhance resilience, increase productivity, and reduce greenhouse gas emissions, CSA provides an integrated pathway for transforming agricultural systems in climate-vulnerable contexts such as Kenya (Waaswa et al., 2024).

The urgency of adopting climate-responsive agricultural approaches is particularly evident in Kenya's arid and semi-arid lands (ASALs), which constitute a significant proportion of the country's land area and are characterized by low and erratic rainfall, land degradation, and resource scarcity (De Jong et al., 2024). Climate change continues to exert pressure on agricultural systems through increased rainfall variability, prolonged droughts, and rising temperatures, all of which affect crop yields, livestock productivity, soil fertility, and water availability (Baarsch et al., 2020; Kebede et al., 2012). Historical evidence further demonstrates that climate variability and land degradation have contributed to declining agricultural productivity over time (Kipkemboi et al., 2021). In response, a wide range of CSA practices, including conservation agriculture, drought-resistant crop varieties, agroforestry, and improved water management, have been promoted by research and development institutions. However, the uptake and scaling of these innovations remain uneven and context dependent across the country.

A critical factor underlying this uneven adoption is the persistent disconnect between research and policy processes. While scientific knowledge on climate-smart interventions has expanded significantly, its translation into coherent and actionable policy frameworks has been inconsistent (Sumberg, 2022). Policy responses are often fragmented across sectors and governance levels, with limited coordination between national priorities and local implementation realities. Furthermore, top-down policy approaches frequently overlook the socio-economic, institutional, and gender dynamics that shape decision-making within agricultural systems, thereby constraining the effectiveness and inclusivity of interventions (Eriksen et al., 2021). These weak feedback loops between research producers and policymakers further limit the ability to adapt policies based on emerging evidence and local realities.

On the other hand, the integration of CSA within the broader Sustainable Development Goals (SDGs) framework further highlights the complexity of aligning agricultural, environmental, and socio-economic objectives. CSA is intrinsically linked to multiple SDGs, particularly those related to food security, climate action, ecosystem sustainability, gender equality, and water resource management. These interconnections create opportunities for generating synergies, such as enhanced resilience, improved livelihoods, and ecosystem sustainability, but also introduce potential trade-offs, including resource competition and uneven distribution of benefits (Nilsson et al., 2016). Despite growing recognition of these dynamics, policy and research frameworks often lack systematic approaches for identifying and managing such interactions.

In Kenya, efforts to mainstream CSA within national and county-level policy frameworks have intensified over the past decade, reflecting increasing awareness of climate risks and the need for coordinated responses. However, important gaps remain. For instance, while policy frameworks are increasingly comprehensive in design, they often exhibit weaknesses in implementation, including limited financing, weak enforcement and accountability mechanisms, and inconsistent integration of evidence into decision-making processes. In addition, the structure of the research landscape, characterized by strong international collaboration but relatively limited local leadership, raises important questions regarding the alignment of research agendas with national development priorities and policy needs.

Against this backdrop, this study examines the policy and research landscape of climate-smart agriculture in Kenya, with a focus on understanding how CSA is integrated within policy frameworks and how it aligns with selected Sustainable Development Goals. Adopting a pattern-based analytical approach, the study seeks to identify systemic trends, institutional dynamics, and recurring gaps in the research–policy interface, rather than undertaking direct comparisons across counties. The overall objective of this study is to strengthen the research–policy interface in climate-smart agriculture by addressing challenges related to coordination, evidence use, and SDG integration. Specifically, the study aims to: assess the current policy and research landscape and its alignment with key SDGs; identify key barriers and gaps in linking research and policy processes; evaluate coordination mechanisms across governance levels; and propose actionable pathways for enhancing policy coherence, strengthening evidence-informed decision-making, and improving integrated governance in climate-smart agriculture.

2. Materials and Methods

2.1 Study Design and Analytical Approach

This study employed a desk-based analytical approach to examine the research–policy interface in Climate-Smart Agriculture (CSA) in Kenya. The design integrates qualitative policy analysis with bibliometric methods to provide a comprehensive assessment of both policy frameworks and research trends. Rather than adopting a comparative county-level framework, the study applies a pattern-based analytical approach to identify systemic trends, institutional characteristics, and recurring gaps across national and subnational policy documents. This approach is particularly suited to studies involving purposively sampled policy materials, where the objective is to derive insights into broader governance dynamics rather than establish direct comparisons across administrative units.

2.2 Policy Document Review and Analysis

A total of 96 national and county-level policy documents were purposively selected for analysis. These included agricultural, environmental, and climate-related strategies, policies, legislative instruments, and planning frameworks, such as the Kenya Climate Smart Agriculture Strategy, the Climate Change Act, and the Agricultural Sector Transformation and Growth Strategy. The selection was guided by relevance to climate change, agriculture, and sustainable development, with the aim of capturing a representative cross-section of policy instruments shaping the CSA landscape in Kenya. Each document was systematically reviewed and coded using a structured analytical framework. Key variables included:

- i. references to Climate-Smart Agriculture (CSA),
- ii. alignment with selected Sustainable Development Goals (SDGs),
- iii. clarity of institutional roles and coordination mechanisms,
- iv. integration of implementation, financing, and monitoring frameworks, and
- v. presence of research–policy linkages and evidence use.

The coding process enabled the development of a policy–SDG analytical matrix, which was used to identify patterns of alignment, gaps, and areas of convergence across the sampled documents. Emphasis was placed on interpreting the implications of these patterns for policy coherence and implementation effectiveness, rather than quantifying differences between specific counties.

2.3 Bibliometric Analysis of CSA Research

To complement the policy analysis, a bibliometric assessment of CSA-related research in Kenya was conducted. A total of 1,307 publications spanning the period 2015–2025 were retrieved from the Web of Science database using relevant search terms associated with climate-smart agriculture. The bibliometric analysis was conducted using the Biblioshiny package in R, which enabled the examination of publication trends, institutional affiliations, and thematic focus areas. Specifically, titles, abstracts, and keywords were analysed to assess the evolution of research priorities, the distribution of institutional contributions, and the extent of collaboration networks within the CSA research landscape. This dual analytical approach i.e., combining policy review and bibliometric analysis, allowed for an integrated understanding of how knowledge is produced and translated into policy within the CSA domain.

2.4 Methodological Considerations and Limitations

This study adopts a desk-based analytical approach to provide a systematic and comprehensive assessment of Kenya’s climate-smart agriculture policy and research landscape. The approach enables broad coverage across national and subnational policy documents and ensures analytical consistency in identifying patterns, institutional dynamics, and alignment with Sustainable Development Goals. The analysis focuses on documented policy and research outputs, capturing systemic trends and governance structures that shape the research–policy interface. The findings therefore reflect overarching policy dynamics rather than context-specific experiences at the local level. This analytical foundation provides a robust basis for informing subsequent empirical research, including stakeholder engagement and field-based validation.

2.5 Contribution and Analytical Scope

This study advances understanding of the research–policy interface in climate-smart agriculture by integrating systematic policy analysis with bibliometric assessment. The combined approach enables a holistic examination of both governance frameworks and knowledge production systems, revealing the structural and institutional factors that shape the effectiveness of CSA implementation in Kenya. The analytical scope focuses on identifying cross-cutting patterns, policy gaps, and alignment challenges across the policy and research landscape. Emphasis is placed on systemic trends rather than case-specific variation, allowing for a broader interpretation of institutional dynamics and policy coherence. The findings are intended to inform evidence-based policy design and support the development of more integrated and accountable approaches to achieving sustainable development outcomes.

3. Results

The results integrate findings from policy analysis and bibliometric assessment to provide a comprehensive understanding of the climate-smart agriculture (CSA) landscape in Kenya. The analysis highlights patterns in policy design, implementation systems, and research production, revealing areas of convergence and misalignment within the research–policy interface.

3.1 Evolution and Integration of CSA in Policy Frameworks

The reviewed policy documents demonstrate a progressively maturing climate governance landscape, with CSA widely embedded across national and county-level frameworks. CSA is referenced in over 97% of documents, reflecting strong alignment with national priorities and global climate agendas (Figure 1).

However, the depth of integration varies. While a substantial proportion of policies explicitly reference CSA, fewer translate this recognition into detailed and actionable measures. The distribution of CSA integration, from implicit references to fully operationalized strategies, reveals a clear gap between policy ambition and implementation readiness (Figure 1).

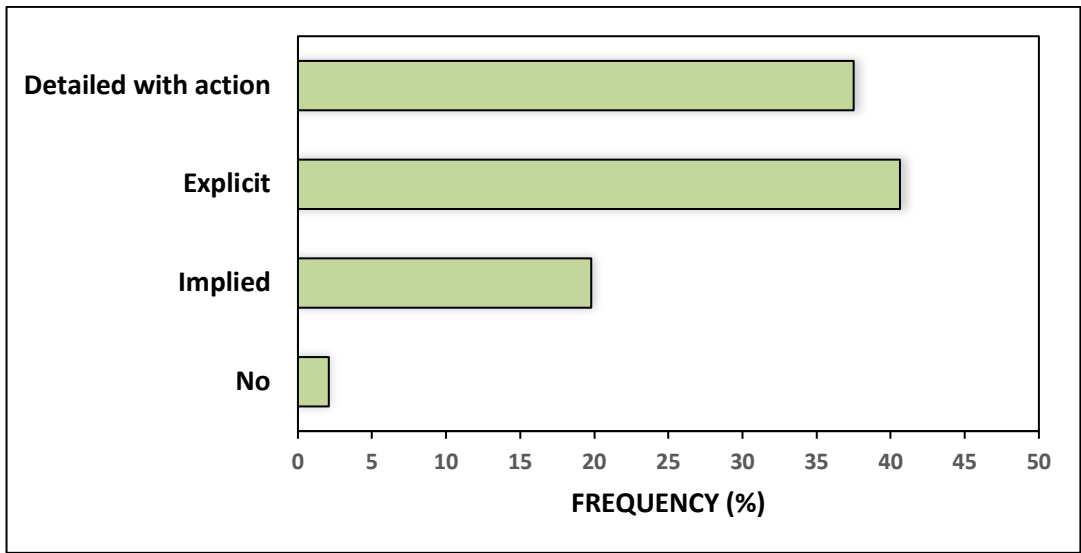


Figure 1: Levels of Climate-Smart Agriculture (CSA) integration across reviewed documents in Kenya.

3.2 Operational Depth and Policy Alignment with Development Goals

The extent to which Climate-Smart Agriculture (CSA) is operationalized within policy documents varies considerably. A substantial proportion of documents move beyond conceptual references to outline concrete interventions. Specifically, 52.1% of the reviewed documents provide detailed descriptions of CSA practices such as sustainable grazing, water management, and early warning systems, while 31.3% go further to include measurable indicators linked to implementation and monitoring (Figure 2) . At the same time, 13.5% of documents remain at a high-level, with limited specification of implementation mechanisms, and 3.1% make no reference to CSA practices at all. This distribution highlights a clear gradient in operational depth, where a significant proportion of policies demonstrate progress toward actionable planning, but gaps remain in translating policy frameworks into consistently implementable and monitorable interventions (Figure 2).

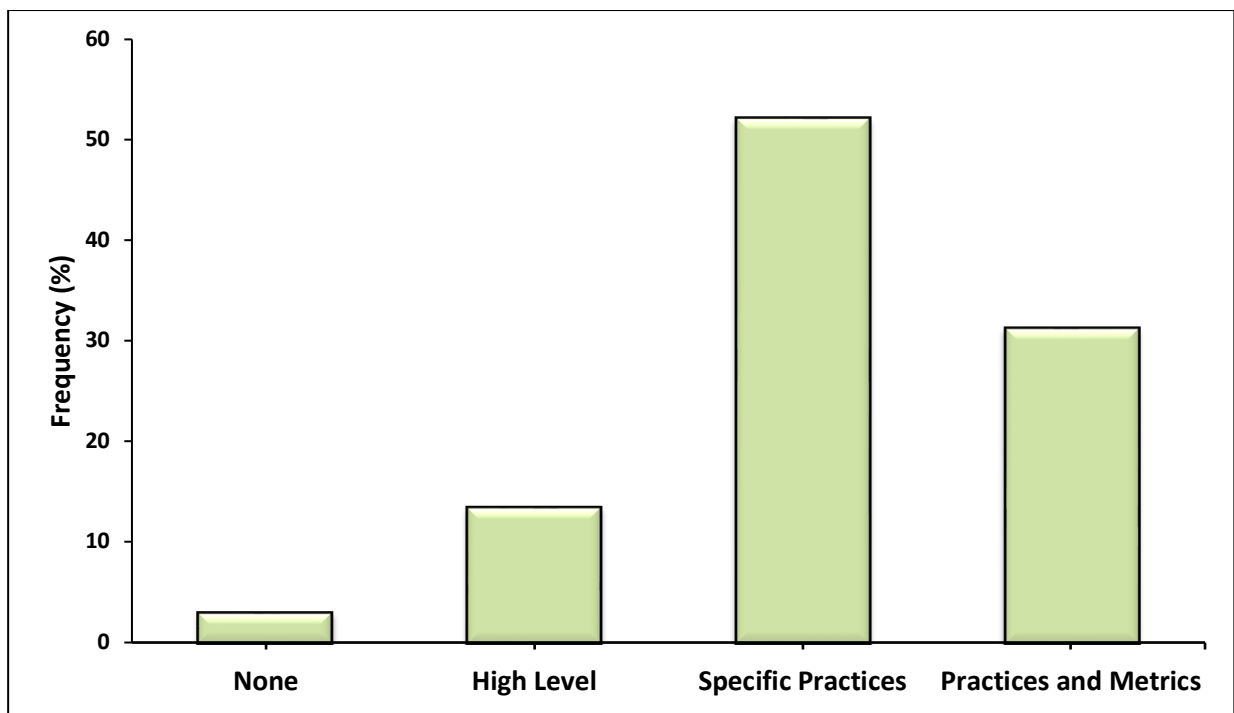


Figure 2: Levels of Climate-Smart Agriculture (CSA) referencing across reviewed documents, showing the extent to which specific practices and metrics were detailed.

Alignment with the Sustainable Development Goals (SDGs) follows a similar pattern. Integration is strongest for SDG 13 (Climate Action), SDG 2 (Zero Hunger), and SDG 6 (Clean Water and Sanitation), which are explicitly linked to actions and indicators in 49%, 41%, and 43% of the documents, respectively (Figure 3). These goals form the core of CSA-related policy design and are consistently embedded within implementation frameworks. In contrast, SDG 5 (Gender Equality) and SDG 15 (Life on Land) are less consistently operationalized. Only 34% and 32% of documents, respectively, include measurable indicators for these goals, while a small proportion (ranging from 1% to 6%) make no reference to them at all. These findings indicate that, although social and ecological dimensions are increasingly acknowledged within policy frameworks, their translation into actionable and measurable components remains limited (Figure 3).

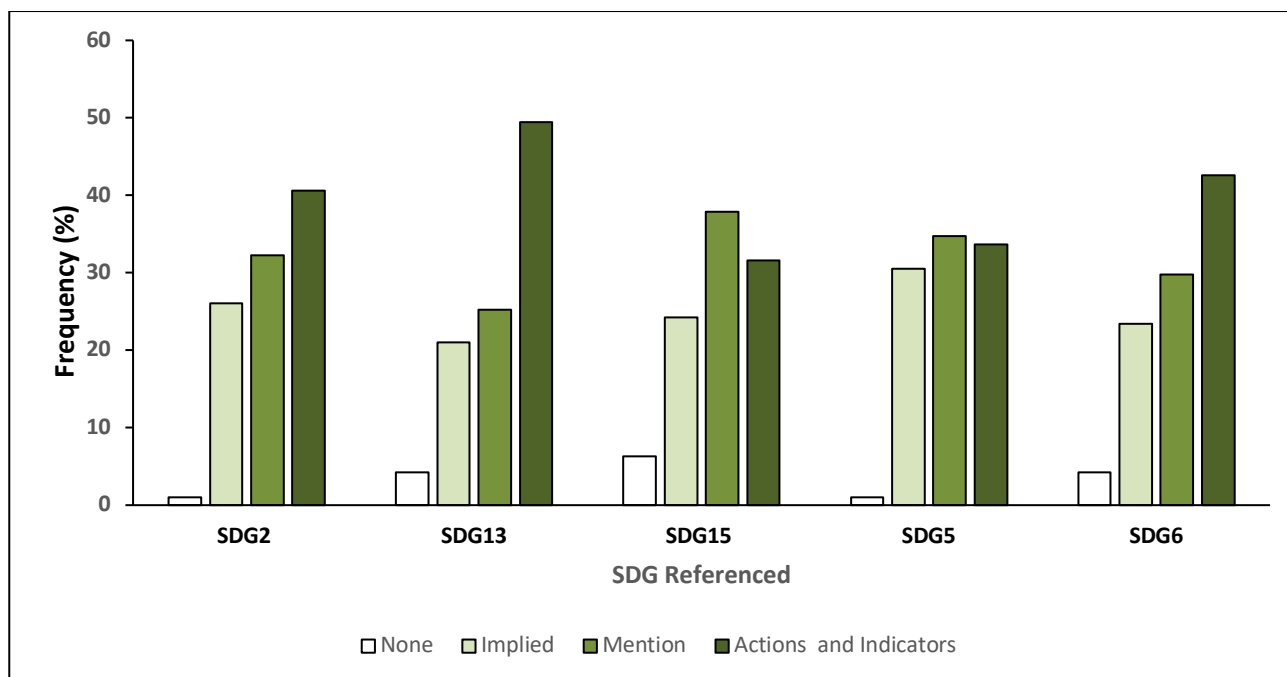


Figure 3: Degree of SDG integration across reviewed policy documents (SDGs 2, 5, 6, 13, and 15).

3.3 Institutional Coordination and Evidence-Policy Linkages

Institutional roles and coordination mechanisms are generally well defined across the reviewed policy documents, indicating a relatively strong governance framework for CSA implementation. Approximately 76.6% of the documents clearly articulate institutional responsibilities and demonstrate coordination across national and subnational levels, reflecting progress in structuring vertical governance systems (Figure 4). A further 14.9% present clearly defined but standalone institutional roles without explicit inter-level linkages, while 6.4% show only partial clarity. A small proportion (2.1%) remains unclear, reflecting fragmented or ambiguous mandates. This distribution suggests that, although institutional design is largely well established, the translation of these arrangements into fully functional coordination systems remains uneven. Gaps persist where coordination mechanisms are either implicit or insufficiently operationalized, limiting effective cross-level implementation (Figure 4).

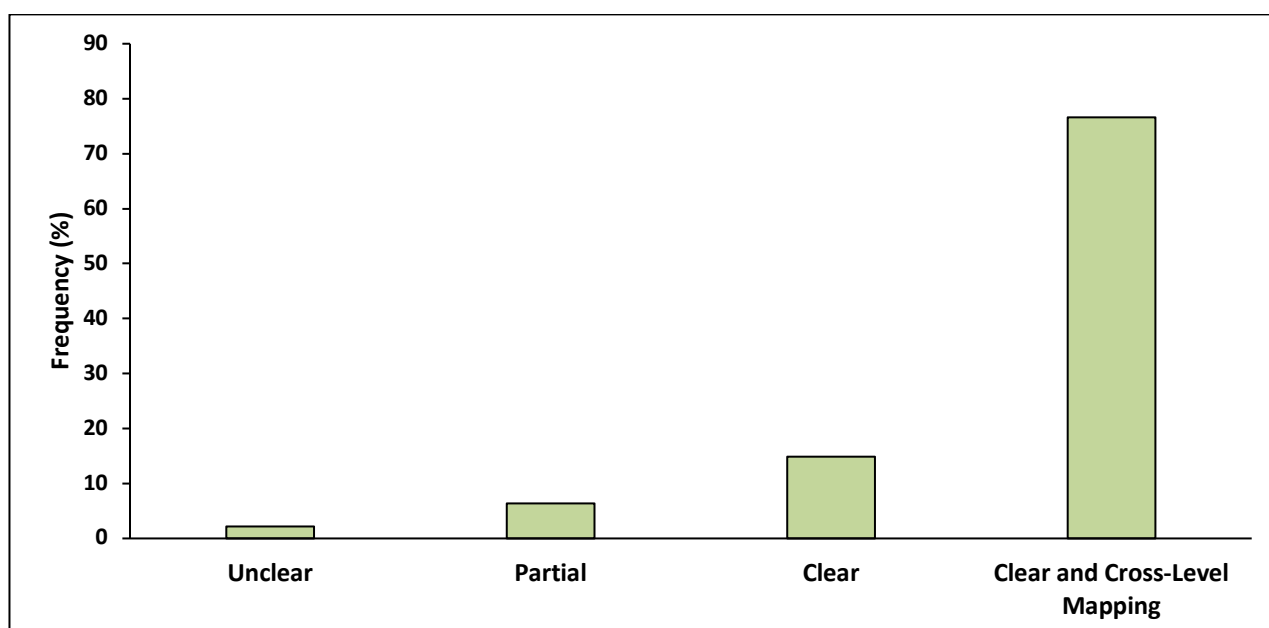


Figure 4: Clarity of institutional responsibilities and vertical coordination across reviewed policy documents.

The integration of evidence into policy processes reflects a similar pattern of partial progress. Approximately 36% of documents explicitly reference evidence-based decision-making, while 29% incorporate formal learning mechanisms such as advisory bodies, technical working groups, or feedback systems (Figure 5). At the same time, 33% of documents rely primarily on anecdotal or general evidence, and 2% make no reference to evidence or learning systems. This distribution indicates that, while there is a growing shift toward evidence-informed policymaking, the institutionalization of systematic evidence use remains uneven. The combined patterns of coordination and evidence integration highlight a critical gap within the research–policy interface. Strong institutional structures exist on paper, but inconsistent use of evidence and weak feedback mechanisms limit the ability of policy systems to adapt, learn, and respond effectively to emerging knowledge (Figure 5).

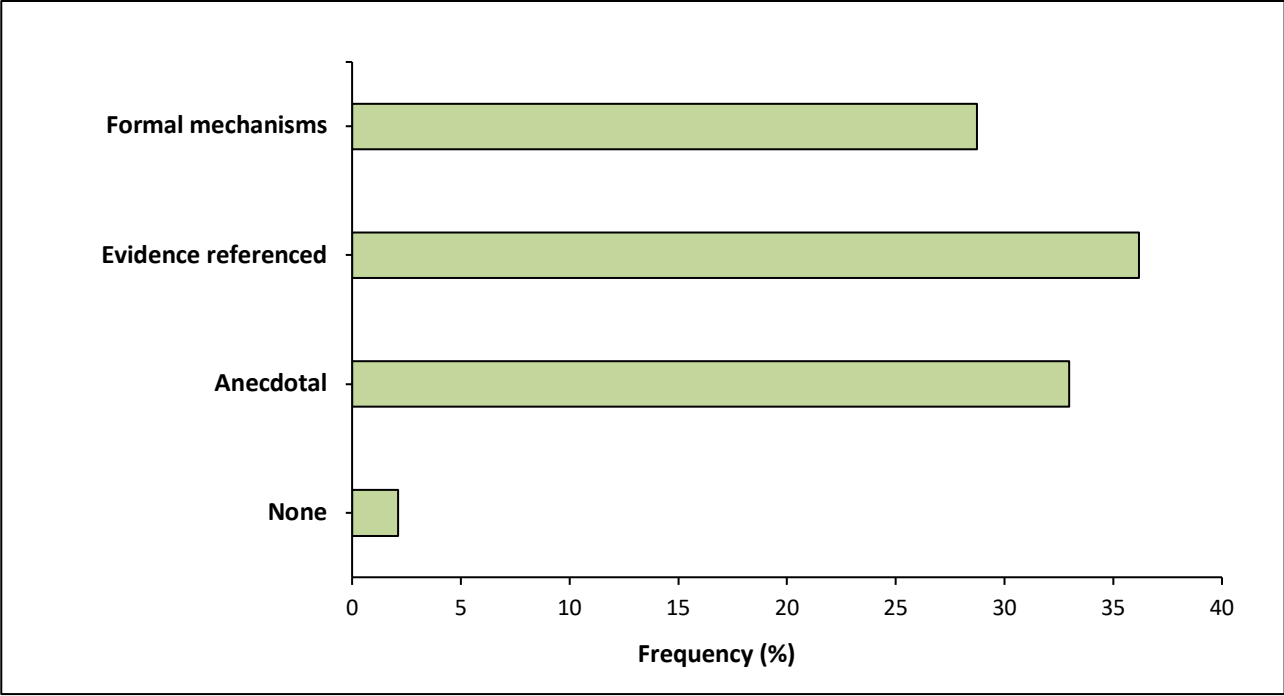


Figure 5: Extent of evidence use and learning mechanisms in reviewed policy documents.

3.4 Implementation Systems: Planning, Financing, and Monitoring

Policy documents demonstrate a strong emphasis on implementation planning, with many outlining structured activities, timelines, and responsible actors. Between 46% and 49% of the reviewed documents present sequenced implementation frameworks with clear milestones and responsibilities, while an additional 33%–35% incorporate dependencies that reflect coordination across sectors and governance levels (Figure 6A). Only 1% of documents lack identifiable implementation structures, and 14%–15% rely on generic arrangements with limited operational detail. These findings indicate strong planning intent across the policy landscape, although execution clarity could be further strengthened through more explicit timelines and accountability mechanisms.

In contrast, financing frameworks are less consistently developed. While 55%–60% of documents reflect an understanding of resource flows and funding linkages, only 20%–22% provide clearly sequenced budgeting frameworks (Figure 6B). A further 14%–15% include only general financial provisions, and 3% make no reference to financing at all. This distribution highlights a critical gap between policy ambition and resource allocation, suggesting that implementation may be constrained by insufficiently defined funding mechanisms. Monitoring and evaluation (M&E) systems are moderately integrated across the reviewed documents. Approximately 39%–41% include detailed indicators linked to implementation dependencies, while 40%–43% outline structured M&E frameworks with defined milestones and targets (Figure 6C). However, 12%–13% rely on generic indicators, and 3% make no reference to M&E systems. Variations in the specificity of baselines, targets, and data sources limit the effectiveness of these frameworks in tracking progress and supporting adaptive management.

Taken together, the comparative distribution of implementation, financing, and monitoring components reveals a clear imbalance within policy systems. Strong planning structures are not consistently matched by equally robust financing and monitoring mechanisms, pointing to a disconnect between policy design and execution capacity (Figure 6).

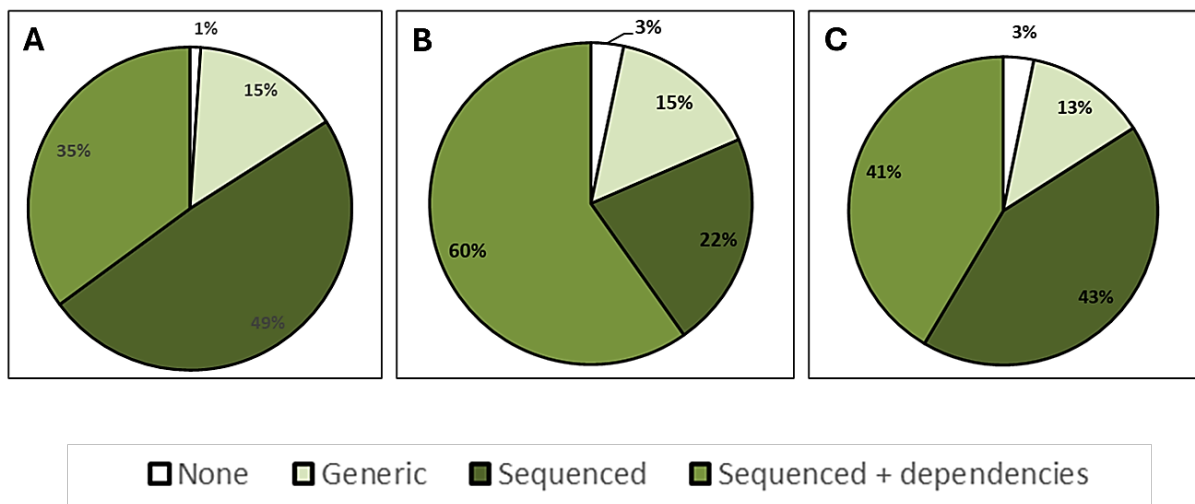


Figure 6: Comparative clarity of implementation (A), financing (B), and monitoring frameworks (C) across reviewed policy documents.

3.5 Trade-offs, Synergies, and Systemic Coherence

Policy documents demonstrate increasing recognition of interactions between Climate-Smart Agriculture (CSA) and broader development objectives, particularly in the form of synergies and co-benefits. A majority of documents (63%) explicitly identify co-benefits, emphasizing integrated outcomes such as improved productivity, enhanced resilience, and environmental sustainability (Figure 7). A further 13% operationalize these synergies through actionable strategies, including cross-sectoral programmes, coordinated implementation mechanisms, and joint financing approaches. Another 17% imply co-benefits without explicit articulation, while only 1% of documents make no reference to synergies at all. These findings indicate a strong policy orientation toward promoting positive interactions across development objectives.

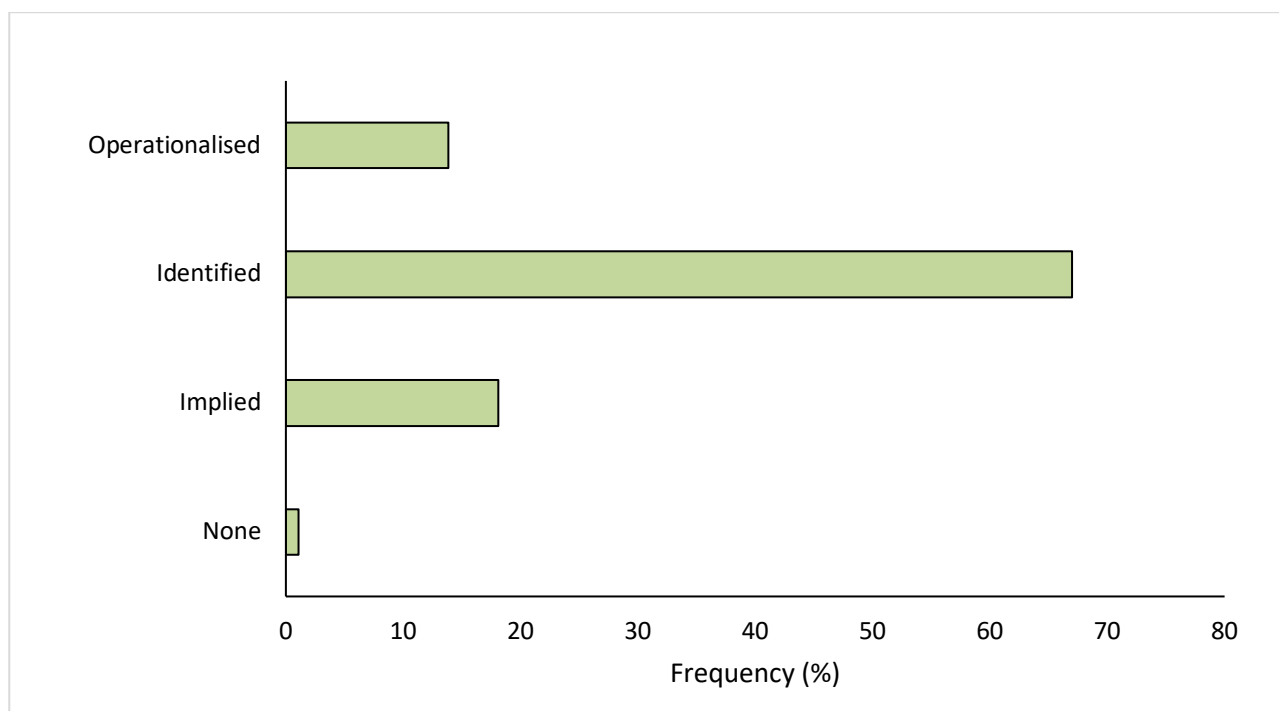


Figure 7: Extent to which policy documents identify and operationalize co-benefits between CSA and SDG objectives.

In contrast, the identification and management of trade-offs remain less developed. Approximately 41% of policy documents explicitly acknowledge trade-offs between CSA and SDG objectives, while only 10% provide concrete mitigation measures to address competing priorities, such as balancing agricultural productivity with biodiversity conservation or gender equity (Figure 8). An additional 37% of documents imply the existence of trade-offs without formal analysis, and 12% make no reference to such interactions. This distribution suggests that, although awareness of potential conflicts is growing, systematic approaches to trade-off assessment and management remain limited within policy frameworks.

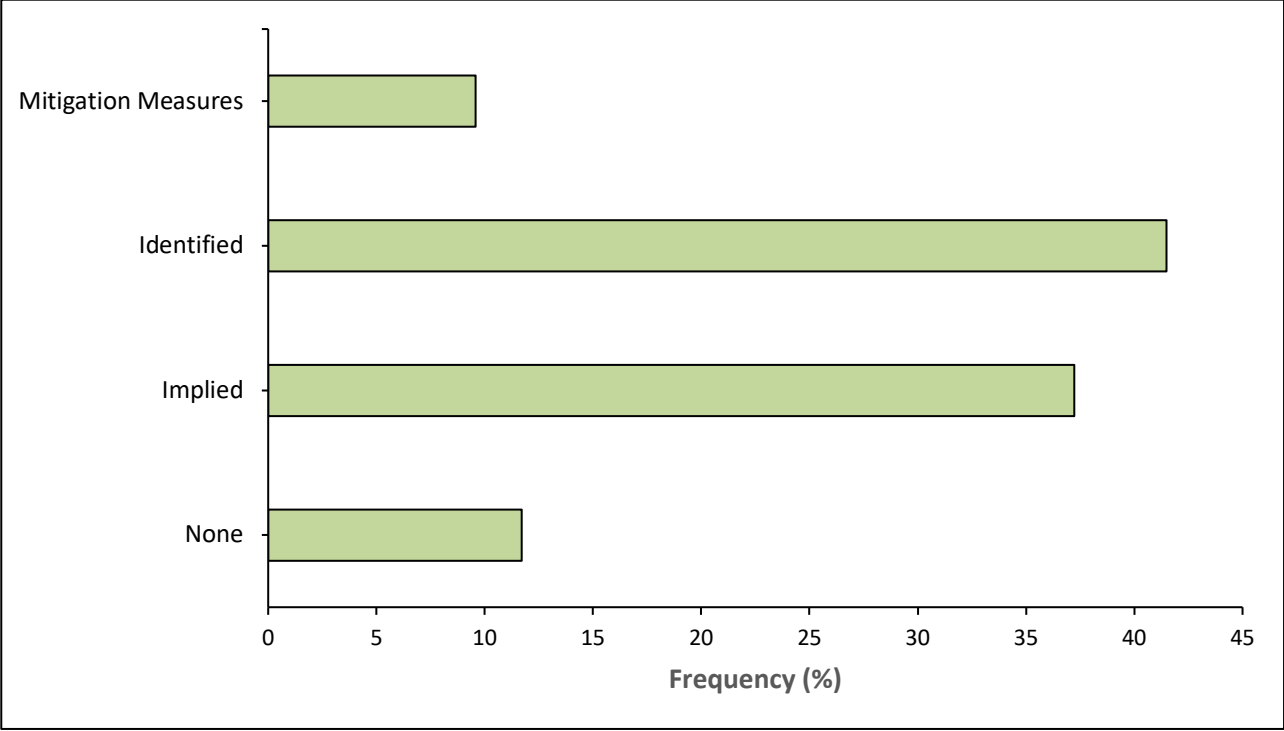


Figure 8: Extent to which policy documents identify and address trade-offs between CSA and SDG objectives.

3.6 Inclusion, Equity, and Indigenous Knowledge Systems

Equity and inclusion are increasingly recognized within climate-smart agriculture (CSA) policy frameworks, particularly in relation to indigenous knowledge systems and vulnerable populations. A majority of the reviewed documents (54%) integrate equity and inclusion considerations within their policy frameworks, reflecting growing attention to participatory and context-specific approaches to climate adaptation (Figure 9). A further 28% of documents go beyond general inclusion to incorporate specific safeguards, including gender-responsive measures, protection of indigenous rights, and participatory planning mechanisms. In contrast, 16% of documents refer to inclusion only in general terms without detailed guidance, while 2% make no reference to equity or indigenous knowledge considerations. These findings indicate significant progress in mainstreaming inclusion within policy design. However, the variation in depth of integration highlights uneven operationalization. While inclusion is widely acknowledged as a principle, the extent to which it is translated into concrete safeguards and implementation mechanisms remains inconsistent, limiting its effectiveness in ensuring equitable participation and benefit-sharing (Figure 9).

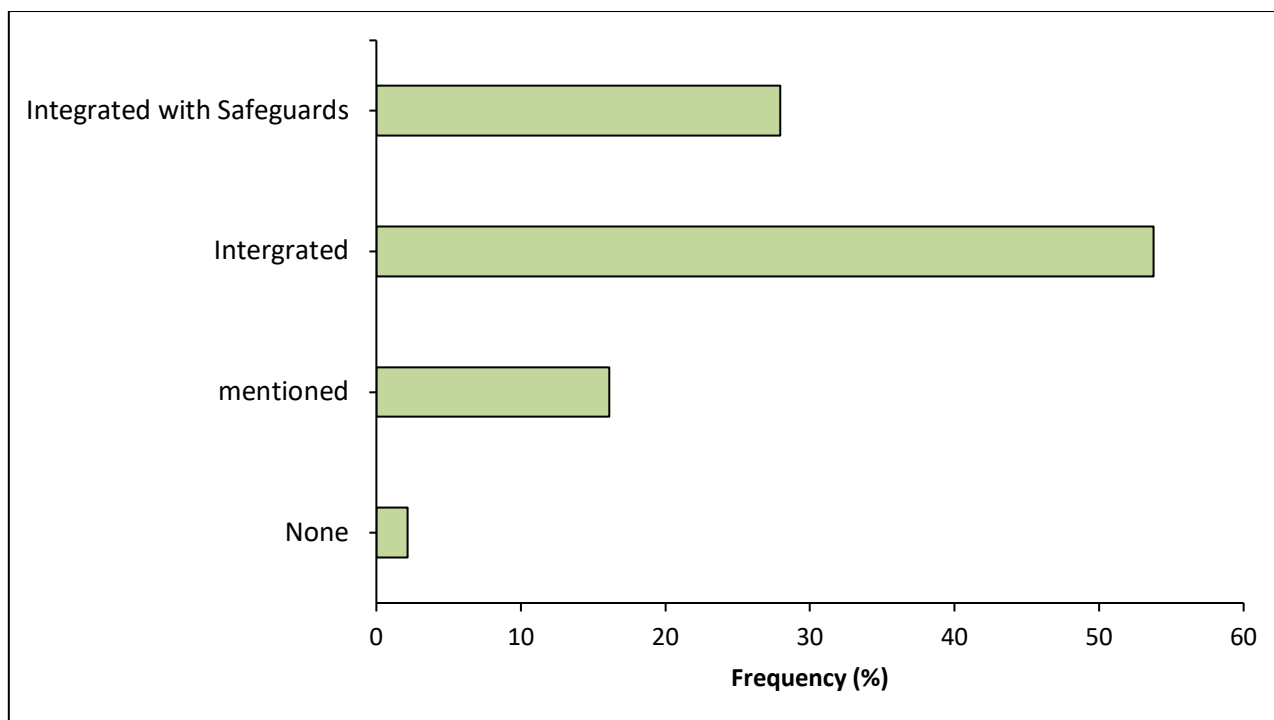


Figure 9: Extent of inclusion of indigenous knowledge and equity safeguards in reviewed policy documents.

3.7 Enforcement, Accountability, and Policy Effectiveness

Enforcement and compliance mechanisms remain among the weakest components of the policy landscape. The reviewed documents show uneven articulation of accountability structures, with only 18% clearly outlining authority mechanisms and proportionate sanctions that define compliance responsibilities (Figure 10). A larger proportion (37%) includes defined powers and sanctions; however, these are often incomplete or weakly specified, limiting their effectiveness in practice. In contrast, 23% of documents provide only minimal or generic commitments, while another 23% make no reference to enforcement provisions at all.

This distribution highlights a significant gap between policy design and enforceability. Although many frameworks recognize the importance of compliance, the absence of clearly defined authority structures, proportionate sanctions, and transparent reporting mechanisms limits their capacity to ensure effective implementation.

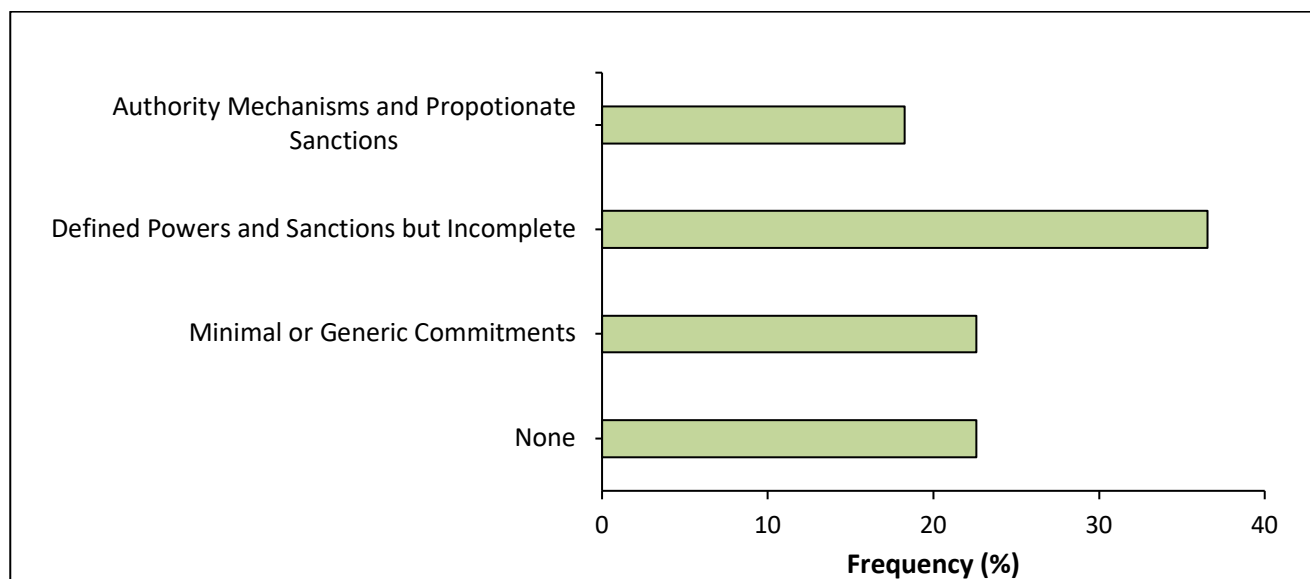


Figure 10: Strength of enforcement and compliance mechanisms across reviewed policy documents.

3.8 Synthesis of Research Trends and Policy Implications

The bibliometric analysis reveals a steady and sustained increase in climate-smart agriculture (CSA) research in Kenya over the past decade, reflecting growing scientific engagement with climate and agricultural challenges. Publication output increased from 22 articles in 2015 to 107 in 2020, followed by accelerated growth to 154 in 2021 and 181 in 2022. Although a slight decline was observed in 2023 (165 publications), output rebounded to 215 publications in both 2024 and 2025, marking the highest levels of research production within the period (Figure 11).

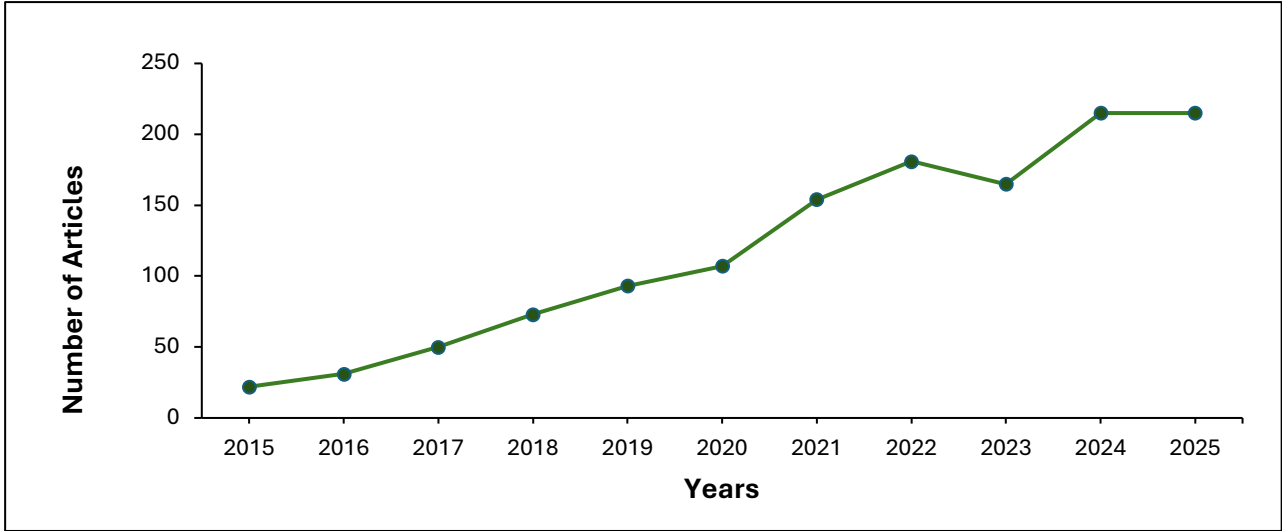


Figure 11: CSA scientific production from 2015-2025

This upward trajectory indicates increasing prioritization of CSA within the research community and provides a strong foundation for evidence-based policymaking and innovation in agricultural systems. Despite this growth, institutional analysis reveals a concentration of research leadership among international organizations. The most prominent contributors include CGIAR centres (292 publications), Wageningen University and Research (110), and the Indian Council of Agricultural Research (91), alongside other major international institutions (Figure 12).

In contrast, Kenyan institutions contribute comparatively fewer publications, with the University of Nairobi (23), Egerton University (19), and Jomo Kenyatta University of Agriculture and Technology (6) representing the most visible local actors. This distribution highlights a structural imbalance in research leadership and suggests a continued reliance on international partnerships for knowledge generation.

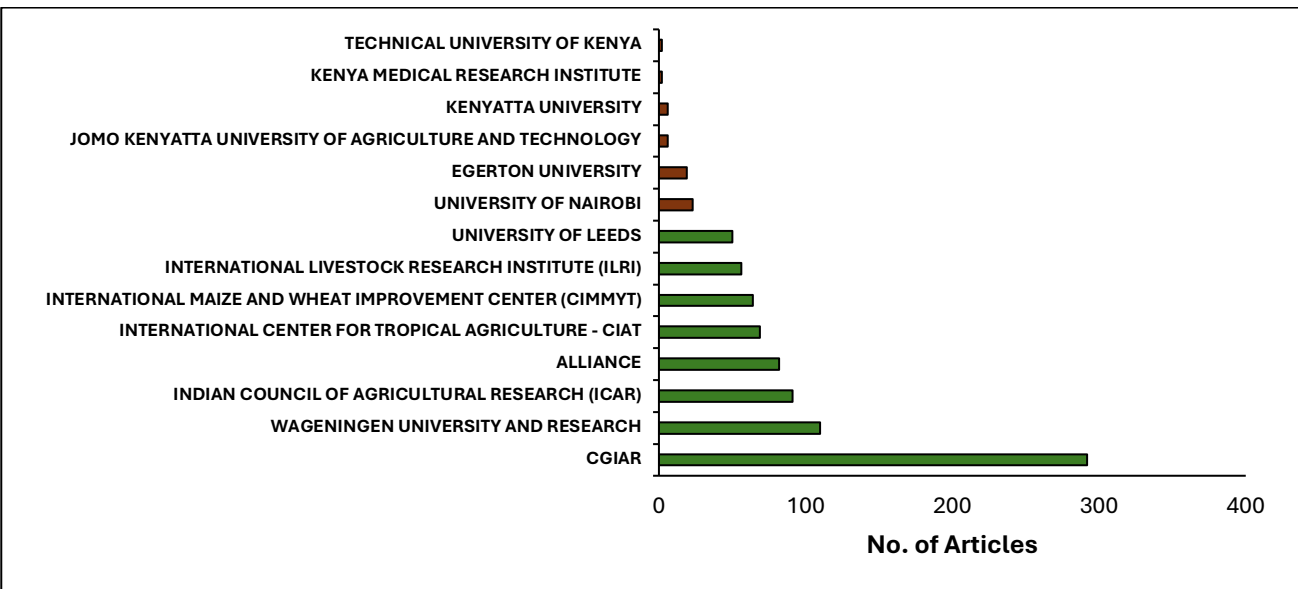


Figure 12: Institutional Affiliations of CSA research in Kenya

Country-level analysis further illustrates the global nature of CSA research linked to Kenya, with the United States (524 publications), India (441), and China (387) leading in output, while Kenya ranks fourth with 245 publications (Figure 13). These patterns indicate strong international collaboration but also raise questions regarding research ownership and agenda-setting.

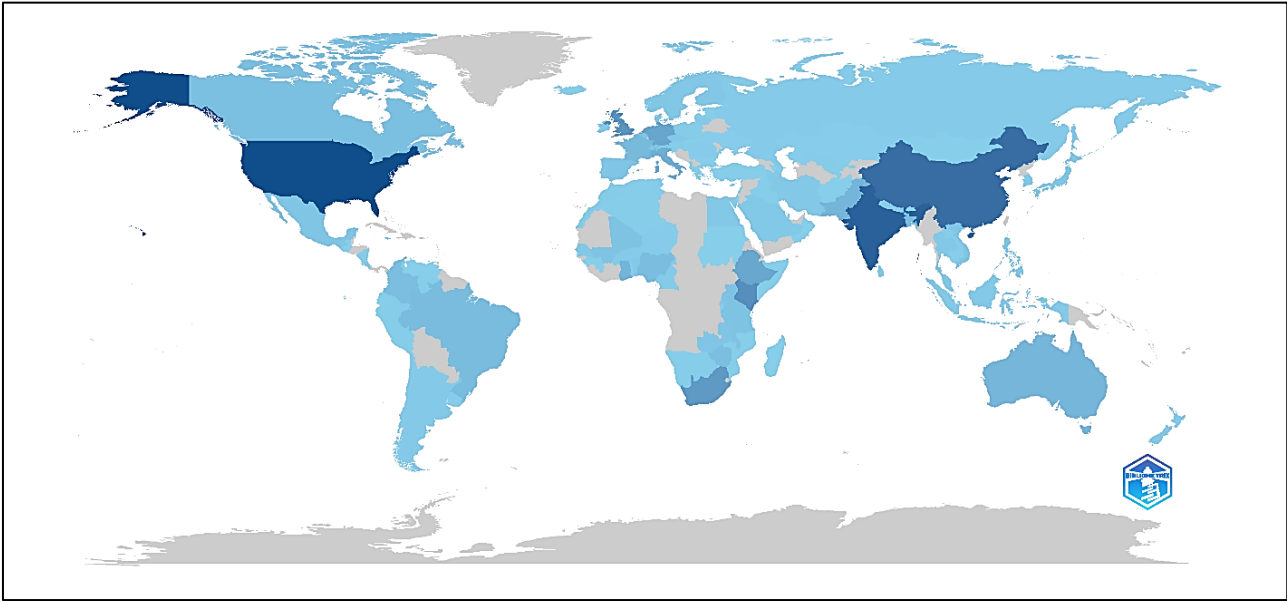


Figure 13: Production per Country for CSA Research in Kenya

In addition, thematic analysis shows that CSA research is heavily concentrated in areas such as adaptation, climate change, food security, and agricultural management. Frequently occurring themes include productivity, soil health, conservation agriculture, and smallholder systems, reflecting a strong focus on resilience and sustainable intensification (Figure 14). In contrast, comparatively less emphasis is placed on areas such as equity, gender, and biodiversity, indicating gaps in the integration of socio-economic and ecological dimensions within the research landscape.

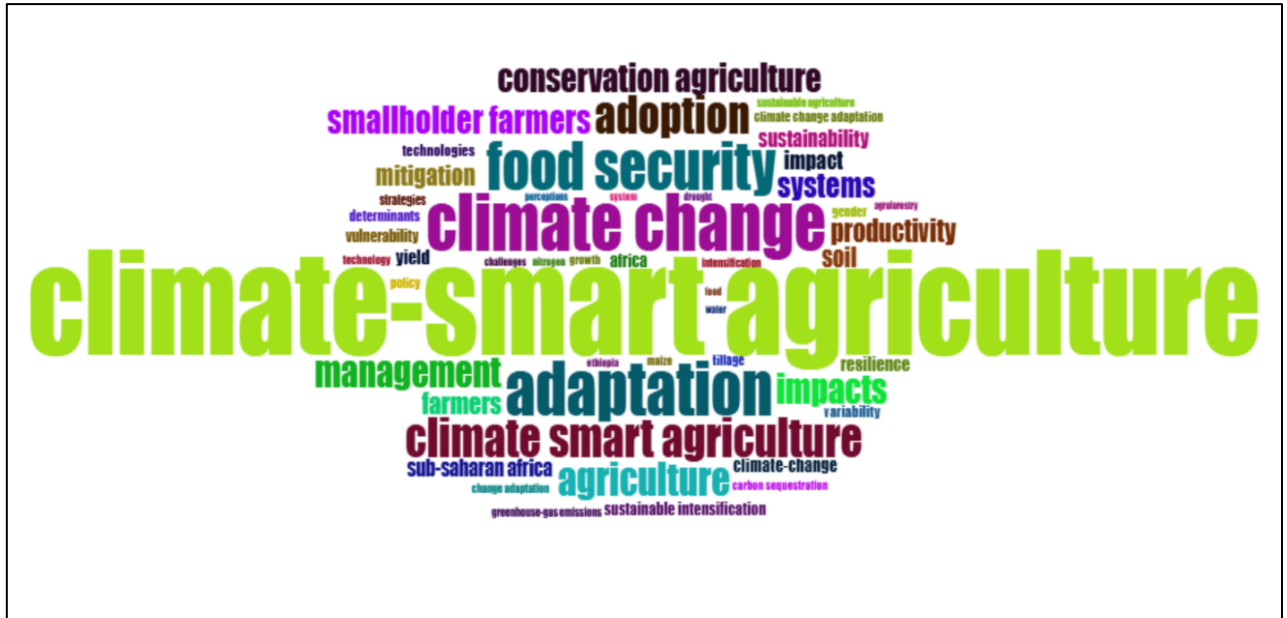


Figure 14: Word Cloud on Thematic focus areas for CSA research in Kenya

4. Discussion

This study examined the research–policy interface in climate-smart agriculture (CSA) in Kenya, focusing on policy alignment, institutional coordination, and implementation systems. The findings reveal a policy landscape that is increasingly aligned with climate and development priorities, yet constrained by gaps in operationalization, financing, enforcement, and evidence integration. These patterns are consistent with broader findings in the literature on climate governance and agricultural transformation in developing contexts.

4.1 From Policy Recognition to Implementation Gaps

The widespread integration of CSA within policy frameworks reflects strong political commitment and alignment with global climate agendas. Similar trends have been observed across Sub-Saharan Africa, where CSA has been rapidly mainstreamed into national strategies (Asfaw & Lipper, 2016; Matteoli et al., 2020). However, as reflected in this study, policy recognition does not consistently translate into implementation capacity. The variation in operational depth across policy documents suggests that while some frameworks are transitioning toward actionable planning, others remain largely aspirational. This pattern points to underlying structural constraints, including differences in technical capacity, institutional readiness, and resource availability. The imbalance between implementation planning, financing, and monitoring further reinforces this gap, indicating that the primary limitation lies not in policy formulation, but in the systems required to support execution.

4.2 Institutional Design versus Functional Coordination

Institutional analysis indicates that governance structures for CSA are generally well defined, with clear mandates across national and subnational levels. This reflects progress in establishing formal coordination systems and aligns with findings from broader studies on climate governance (Eriksen et al., 2021). However, the persistence of coordination gaps suggests that institutional design does not necessarily translate into functional coordination. In many cases, coordination mechanisms remain procedural, with limited interaction, weak accountability linkages, and insufficient feedback systems. Similar challenges have been identified in multi-level governance systems, where institutional fragmentation limits the effectiveness of policy implementation (Nilsson et al., 2016). These findings highlight the distinction between institutional presence and institutional performance, suggesting that coordination challenges are rooted in how systems operate, rather than how they are structured.

4.3 Evidence Use and the Research-Policy Disconnect

The uneven integration of evidence into policy processes reflects a persistent gap within the research–policy interface. Although some policy documents demonstrate the use of scientific inputs and structured learning mechanisms, reliance on general or anecdotal evidence remains significant. This pattern is consistent with existing literature, which identifies challenges in translating scientific knowledge into policy-relevant outputs (Turnheim et al., 2020). The issue is not the absence of research, but the lack of institutional mechanisms to embed evidence within policymaking processes. Weak feedback loops further constrain the ability of policies to adapt in response to emerging knowledge. The findings suggest that knowledge production and policy application operate in parallel rather than as an integrated system, limiting the effectiveness of evidence-informed decision-making.

4.4 Donor Dominance and Its Implications for Policy Relevance

The bibliometric analysis reveals a strong and sustained increase in CSA research output, indicating growing scientific engagement. However, the concentration of research leadership among international institutions reflects a donor-driven knowledge production system, a pattern widely observed in African research contexts (Jayne et al., 2023). This dynamic has important implications for policy relevance. Research agendas shaped by external actors tend to prioritize globally recognized themes, such as adaptation and productivity, while underemphasizing context-specific challenges related to governance, financing, and social equity. As a result, the available evidence base may not fully align with national policy priorities or implementation needs. The findings suggest that donor dominance influences not only the volume of research, but also its thematic focus and applicability within local policy contexts.

4.5 Implications for Kenya's Research and Policy Priorities

The relatively lower contribution of Kenyan institutions to CSA research highlights structural constraints within the national research ecosystem. Similar patterns have been documented in studies on research capacity in developing countries, where limited funding, infrastructure, and institutional support constrain local leadership. This imbalance affects the alignment between research outputs and national priorities. Locally led research is more likely to reflect contextual realities, including socio-economic conditions and institutional constraints, making it more relevant for policy design and implementation. The findings therefore underscore the importance of strengthening national research systems to enhance both the relevance and impact of CSA-related knowledge.

4.6 Trade-offs, Synergies, and Policy Complexity

The strong emphasis on synergies, combined with limited attention to trade-offs, reflects a broader tendency in policy frameworks to prioritize positive outcomes while under-addressing complexity. Previous studies have emphasized the importance of explicitly identifying and managing trade-offs within SDG frameworks to avoid unintended consequences (Nilsson et al., 2016).

The limited operationalization of trade-off management observed in this study suggests that current policy frameworks are not yet fully equipped to handle competing objectives. This is particularly relevant in areas such as land use, biodiversity conservation, and social equity, where policy decisions often involve complex trade-offs. The findings indicate that while integrated thinking is emerging, it remains incomplete in its application.

4.7 Inclusion, Equity, and the Limits of Mainstreaming

The increasing recognition of equity, gender, and indigenous knowledge within policy frameworks reflects progress toward inclusive climate governance. However, the variation in operational depth suggests that mainstreaming remains uneven. In many cases, inclusion is articulated as a guiding principle but not consistently translated into concrete safeguards or implementation mechanisms. This pattern is consistent with findings in climate adaptation literature, where inclusion is often conceptually acknowledged but weakly operationalized (Eriksen et al., 2021). The limited integration of these dimensions within both policy and research systems highlights a gap in addressing the social aspects of climate-smart agriculture.

4.8 Enforcement as the Critical Missing Link

The findings identify enforcement and accountability as the weakest components of the policy system. The absence of clearly defined authority structures, proportionate sanctions, and compliance mechanisms limits the effectiveness of policy frameworks. Without enforcement mechanisms, policy commitments remain largely aspirational, regardless of their design quality. The results suggest that strengthening accountability systems is central to improving the effectiveness of CSA implementation.

4.9 Synthesis: Understanding the Research–Policy Interface

Taken together, the findings point to a policy and research system that is structurally well developed but functionally uneven. Policy frameworks demonstrate strong alignment with climate and development goals, while research output is expanding. However, persistent gaps in coordination, evidence use, financing, enforcement, and inclusion limit overall effectiveness. These gaps are interconnected and reflect broader systemic challenges within the research–policy interface. Addressing them requires a shift from fragmented approaches toward more integrated systems that align knowledge production, policy design, and implementation processes.

5. Conclusion and Policy Implications

Kenya has made significant progress in integrating Climate-Smart Agriculture (CSA) into national and county-level policy frameworks, reflecting increasing recognition of agriculture's role in sustainable development and climate resilience. The widespread inclusion of CSA, alongside growing alignment with key Sustainable Development Goals, demonstrates a maturing policy landscape that is responsive to climate and development challenges.

However, the findings reveal that this progress is uneven. While policy frameworks are generally well developed and institutionally structured, gaps persist in implementation systems, particularly in financing, enforcement, and evidence integration. The imbalance between strong planning structures and weaker accountability and resource mechanisms limits the translation of policy commitments into measurable outcomes. At the same time, although equity, indigenous knowledge, and SDG interactions are increasingly recognized, their operationalization remains inconsistent across policy instruments.

The research landscape reflects similar patterns. While CSA-related scientific output has grown substantially, knowledge production remains largely driven by international institutions. This donor-dominated structure raises important questions regarding research ownership, agenda-setting, and alignment with national priorities. As a result, critical areas such as governance, financing, and socio-economic dimensions remain underrepresented in both research and policy systems.

Addressing these challenges requires a shift from policy recognition to system functionality. Strengthening implementation systems, improving coordination, and aligning research with policy needs will be central to enhancing the effectiveness of CSA in Kenya.

Policy Implications

The following priority actions emerge from the analysis:

1. **Strengthen implementation and accountability systems:** Develop clear enforcement frameworks, including defined institutional mandates, proportionate sanctions, and transparent reporting mechanisms to ensure policy compliance and effectiveness.
2. **Enhance financing mechanisms for CSA:** Integrate explicit budgeting frameworks within policy design and expand sustainable financing options, including climate funds, blended finance, and public–private partnerships.
3. **Institutionalize the research–policy interface:** Establish formal platforms for knowledge exchange, including advisory bodies, policy–research forums, and structured feedback mechanisms to support evidence-informed decision-making.
4. **Strengthen local research capacity and leadership:** Increase investment in national research institutions and promote locally led research to ensure alignment with Kenya’s development priorities and contextual realities.
5. **Integrate trade-off analysis into policy frameworks:** Adopt systems-based approaches, including multi-criteria analysis and scenario planning, to better manage competing objectives across sectors.
6. **Advance equity and inclusive policy design:** Move beyond conceptual recognition by embedding gender-responsive indicators, participatory mechanisms, and safeguards for indigenous knowledge systems within implementation frameworks.
7. **Improve coordination across governance levels:** Strengthen inter-agency collaboration through joint planning, integrated monitoring systems, and clearly defined accountability structures across national and county levels.

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